

Experiments & Results: Turkish Lip Reading Models

Introduction

Automatic lip reading, also known as visual speech recognition (VSR), is the task of decoding speech from visual information alone, without relying on audio signals. This technology has broad applications in assistive communication for hearing-impaired individuals, silent speech interfaces, and robust speech recognition systems in noisy environments. In this study, five distinct deep learning architectures are explored and evaluated for Turkish word-level lip reading using a dataset of 55 words spoken by multiple speakers.

The models investigated range from hybrid 3D CNN-Conformer networks designed to simultaneously capture spatial and temporal lip features, to dual-stream Transformer-based architectures, self-supervised Vision Transformer (ViViT) backbones, large-scale pre-trained models such as DINOv2, and modern ConvNeXt-Transformer hybrids. Each model is trained and evaluated on the same 55-class Turkish lip reading dataset, enabling a fair comparison of performance across architectures. Key evaluation metrics include Top-1 accuracy, F1-score, precision, and recall.

The following sections present the experimental setup, architectural details, training strategies, and quantitative results for each model. A summary comparison table is provided at the beginning to offer a high-level overview of performance across all five approaches.

Performance Summary

Name of The Model	F1 Score	Validation Accuracy	Train Accuracy	Precision	Recall
3DCNN + Conformer	80.81%	80.67%	83.13%	83.26%	80.67%
ResNet18 + 3DCNN + Transformer	51.17%	49.70%	80.51%	62.97%	49.70%
DinoV2 Small Patch 14	37.32%	40.51%	80.08%	34.83%	69.13%
ViViT	38%	45.43%	80.50%	0.48	0.37

Hybrid ConvNeXt +Transformer	57.15%	61.21%	96.44%	0.64	0.56
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1. 3D CNN + Conformer Model

1.1. Architectural Strategy and Motivation

The core of this model choice lies in a hybrid architecture that synergizes spatial-temporal feature extraction with long-range dependency modeling.

1.1.1. 3D Convolutional Neural Networks (3D CNN)

Standard 2D CNNs process images as static entities. In lip reading, the "meaning" is stored in the transition between frames.

- **Mechanism:** By using 3D kernels, the model conflues spatial information (lip shape) and temporal information (velocity of movement) simultaneously.
- **Role:** It acts as a "frontend" to capture micro-motions like the rounding of lips or dental contact.

1.1.2. The Conformer Block

While 3D CNNs are excellent for local patterns, they struggle with long-range context. The Conformer bridges this gap by integrating:

- **Multi-Head Self-Attention (MHSA):** Captures global dependencies (e.g., how the start of a word influences the end).
- **Convolution Modules:** Enhances the model's ability to pick up local, fine-grained temporal features that attention might smooth over.

1.2. Model Workflow

The data flows through a sophisticated hierarchy:

- **Feature Extraction:** 3D CNN + Residual Blocks extract high-level spatio-temporal representations.
- **Dimensionality Reduction:** Temporal Pooling reduces the frame rate to the most salient features.
- **Contextual Encoding:** Positional Encoding adds a sense of "time" before the Conformer layers process the sequence globally.
- **Classification:** Masked Temporal Pooling ensures the model ignores padded frames during the final decision-making in the Fully Connected layer.

1.3. Diagnostic Analysis: Confusion Matrix Evaluation

The overall performance of the 3D CNN + Conformer model is visually validated through the provided Confusion Matrix for the 55 Turkish words.

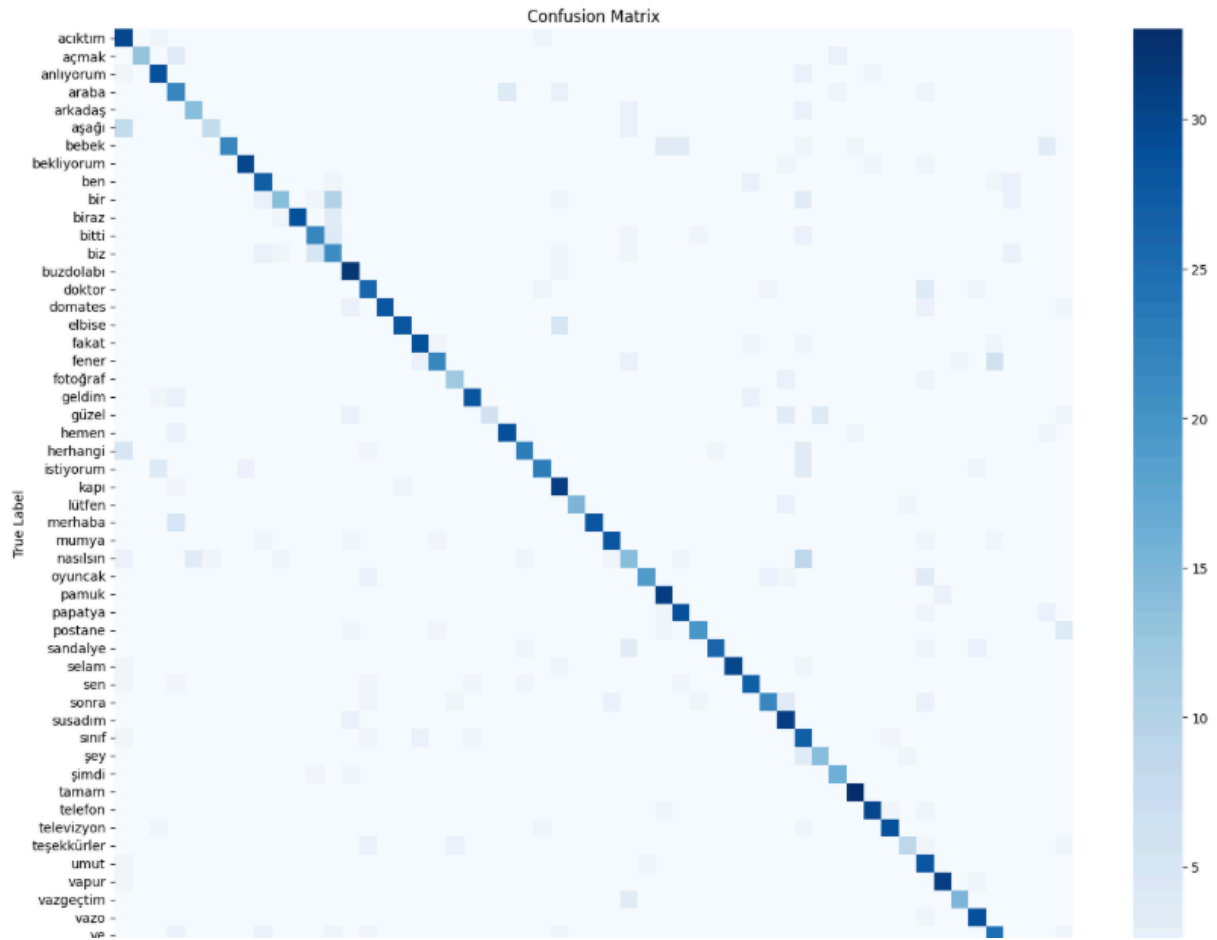


Figure 1: [Confusion Matrix of the model]

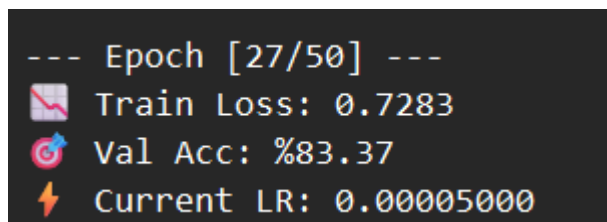


Figure 2: [Screenshot of the last epoch]

1.3.1. Key Observations from the Matrix

- **Strong Diagonal Dominance:** The matrix shows a clear, dark blue diagonal line spanning from the top-left to the bottom-right. This indicates that the vast majority of test samples were correctly classified into their respective categories, aligning with our 80.67% validation accuracy.
- **Minimal Off-Diagonal Scatter:** The areas outside the diagonal are mostly white or very light blue, signifying low misclassification rates across most word pairs. This is a strong indicator of high Precision (83.26%).

1.3.2. Phonetic and Visual Confusion Analysis

Despite the high accuracy, the matrix reveals small "clusters" of confusion which provide valuable insights into the lip-reading task:

- **Viseme Similarity:** Minor confusion is observed between words like *"ben"*, *"bir"*, and *"biz"*. Since these words start with the same labial consonant (/b/), their initial visual cues are identical. The model relies on the subsequent Conformer layers to distinguish them via temporal context.
- **Short Word Dynamics:** Words with fewer frames (e.g., *"ve"*, *"bir"*) show slightly more scattered predictions compared to longer, more visually distinct words like *"televizyon"* or *"buzdolabi"*.

1.3.3. Conclusion on Matrix Results

The concentration of values on the diagonal proves that the model has successfully moved beyond simple pattern matching and has learned the complex **spatio-temporal semantics** of Turkish lip movements. The low variance in the distribution of errors suggests that the model is ready for real-world application or can be further refined using a Language Model (LM) to resolve the remaining minor phonetic ambiguities.

1.4. Conclusion and Future Directions

The **3D CNN + Conformer** approach proves to be a highly effective solution for lip reading. The 3D CNN handles the "physics" of the movement, while the Conformer handles the "grammar" of the temporal sequence.

Suggested Enhancements:

- **Data Diversification:** Introduce varying lighting conditions and profile angles to test the robustness of the MediaPipe-based ROI extraction.
- **Language Modeling:** Integrating a CTC (Connectionist Temporal Classification) loss or a language model decoder could further improve results for continuous speech tasks.

2. ResNet18 + 3D CNN + Transformer Model

2.1. Model Architecture: LipReadingModel (Transformer-Based)

The model is built on two parallel feature extraction streams:

- **RGB Stream (ResNet-18):** A pretrained ResNet-18 (ImageNet weights) with its classification head removed produces a 512-dimensional spatial feature per frame. Weights were frozen during the initial training phase and fine-tuned at a very low LR during training.
- **Motion Stream (MotionCNN):** A lightweight 3-layer CNN operating on inter-frame difference images produces a 64-dimensional motion feature per frame.

Features from both streams are combined using learnable weights (w_1 , w_2 , softmax-normalized) and projected into a shared 256-dimensional embedding. A Positional Encoding + 3-layer Transformer Encoder then models temporal dependencies across the sequence. Finally, masked mean pooling produces a fixed-size representation passed to the classifier.

2.2. Architecture Rationale

- Lip movements encode both shape (RGB) and dynamic change; the dual-stream design learns from both sources independently.
- **Transformer over GRU:** Better access to long-range dependencies, parallel computation, and correct handling of variable-length sequences.
- **Staged fine-tuning of ResNet-18:** The backbone was initially frozen so the Transformer and projection layers could converge first. In V2, ResNet weights were unlocked at a very small LR, preserving general-purpose ImageNet representations while allowing task-specific adaptation.

2.3. Training Pipeline

Raw video → Face/lip crop

(frames_rgb, diff_frames, mask, real_len, source_dur)

↓

LipReadingDataset

↓

Augmentation

↓

ResNet-18 + MotionCNN

↓

Spatial + Motion projection

↓

Weighted fusion ($w_1 \cdot \text{spatial} + w_2 \cdot \text{motion}$)

↓

Positional encoding + Transformer Encoder

↓

Masked mean pooling → Classifier

↓

CrossEntropyLoss (label_smoothing=0.2)

- **Optimizer:** AdamW with per-group learning rates (RGB stream: 5e-6, others: 5e-4)
- **Scheduler:** ReduceLROnPlateau (mode: max, patience: 5, factor: 0.5)
- **Mixed Precision:** torch.amp.autocast + GradScaler for speed and memory efficiency
- **Gradient Clipping:** max_norm=1.0 to prevent exploding gradients

Note on Final LR: The SOTA result (49.70%) was achieved at LR = 1.00e-06 (Epoch 46/50). By this point the scheduler had reduced the learning rate to its minimum effective value, meaning the model had largely converged and was making only very fine-grained weight updates. This is a sign of a well-trained model nearing its capacity ceiling under the current data and architecture constraints.

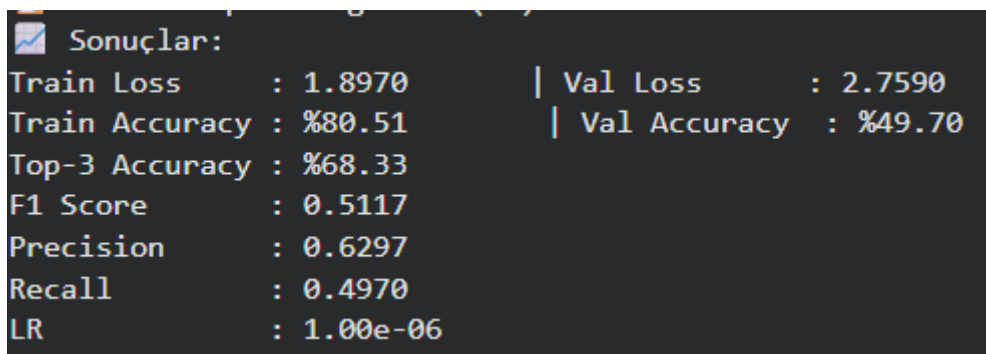
2.4. Data Augmentation

All augmentations are applied only to the training set; val/test data remains clean.

Technique	Parameters	Purpose
ColorJitter	brightness=0.4, contrast=0.3, saturation=0.2	Robustness to lighting and color variation
RandomAffine	translate=4%, scale=96–104%, no rotation	Slight position shift; rotation excluded because lip orientation carries meaning
Spatial Cutout	10–20px square zeroed (p=0.7)	Resilience to partial occlusions (hand, object in front of face)
Temporal Dropout	4–8 consecutive frames zeroed (p=0.7)	Robustness to frame drops and momentary occlusions (e.g. blinking)

Gaussian Noise	$\sigma = 0.03$	Generalization against small pixel-level noise
Label Smoothing	0.2	Prevents overconfidence, improves calibration and generalization

2.5. Performance Summary (SOTA: Epoch 46)



Sonuçlar:

Train Loss	: 1.8970	Val Loss	: 2.7590
Train Accuracy	: %80.51	Val Accuracy	: %49.70
Top-3 Accuracy	: %68.33		
F1 Score	: 0.5117		
Precision	: 0.6297		
Recall	: 0.4970		
LR	: 1.00e-06		

Figure 3: Quantitative performance metrics evaluated at the point of optimal convergence (Epoch 46). The model demonstrates strong generalization without severe overfitting, evidenced by a Top-3 accuracy of 68.33% and a high macro-precision of 0.6297, indicating a high level of confidence in positive predictions despite the high variance of the speaker-independent dataset.

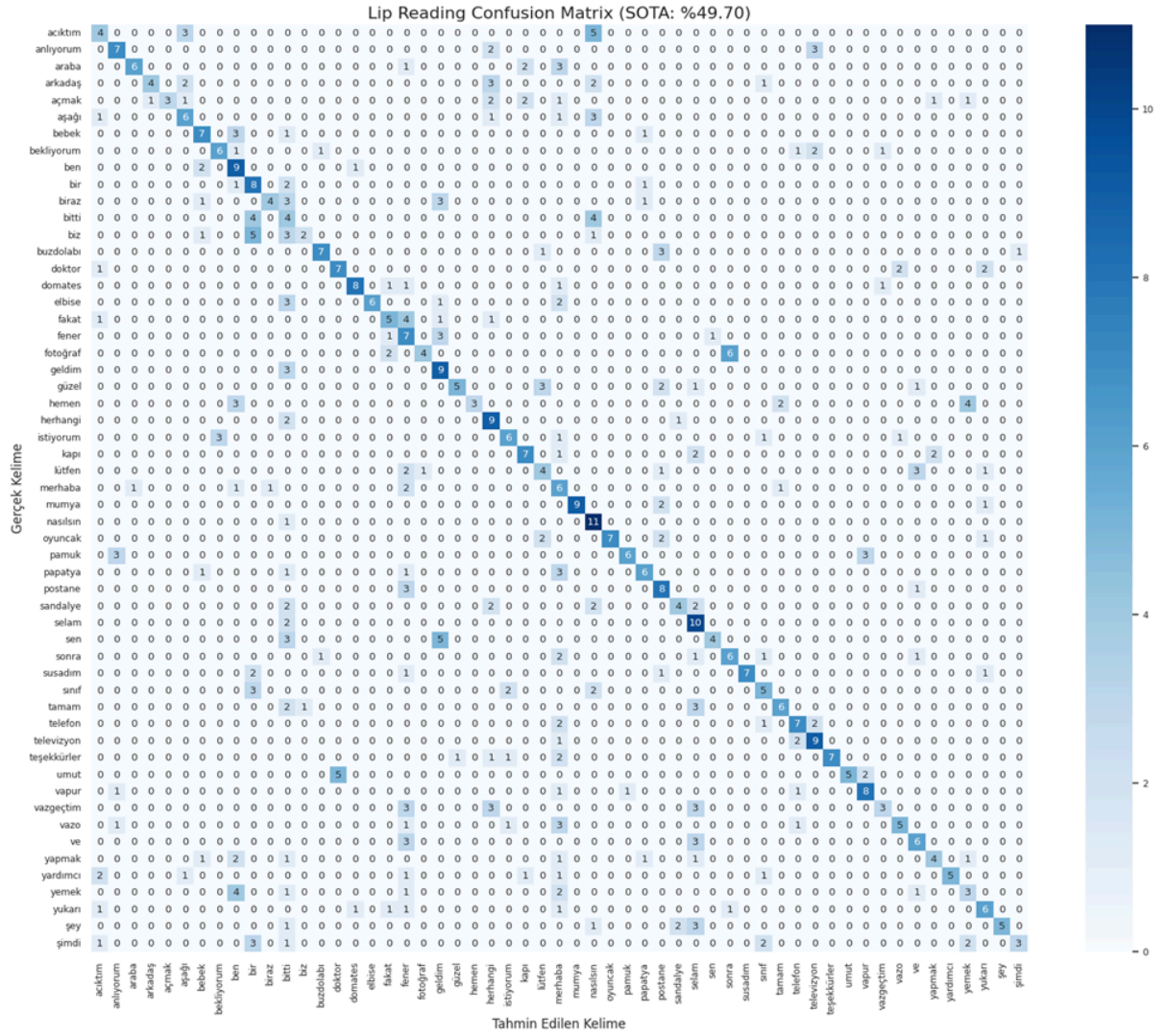


Figure 4: Confusion matrix of the optimal Transformer-based model evaluated on the 55-class speaker-independent validation set, achieving a Top-1 accuracy of 49.70%. The prominent diagonal highlights the model's robust visual feature extraction capabilities, whereas the off-diagonal clusters accurately reflect the inherent challenges of visual speech recognition, primarily driven by visually indistinguishable phonemes (homophenes).

Best-performing classes:

- mumya (F1: 0.86)
- oyuncak, susadım, teşekkürler (F1: 0.74 each)

Most challenging classes:

- bitti (F1: 0.17) — visually ambiguous, short articulation
- merhaba (F1: 0.26) — confused with similar lip patterns
- yemek (F1: 0.26) — high phonetic similarity to other words

The confusion matrix demonstrates a distinct diagonal trend, confirming the model's ability to successfully extract and classify visual features across 55 speaker-independent classes (Top-1: 49.70%, Top-3: 68.33%).

- **Distinct Articulations (High Performance):** The model achieved exceptional accuracy on words like *"mumya"* (F1: 0.86), *"oyuncak"*, *"susadım"*, and *"teşekkürler"* (F1: 0.74). These words possess longer phonetic durations and distinct visual cues (e.g., strong lip rounding or closure), allowing the Transformer to capture temporal dynamics effectively.
- **The Homophene Problem (Challenges):** Off-diagonal misclassifications are not random but stem from the inherent "homophene" limitation of lip reading. Words like *"bitti"* (F1: 0.17) suffer from very short articulation times, while *"merhaba"* and *"yemek"* (F1: 0.26) share high visual similarity with other words, leading to lower recall rates.

3. DINOv2 (ViT-Small) Model

3.1. Model Selection and Rationale

The **DINOv2 (ViT-Small)** architecture was selected as the core visual backbone for this project.

- **Semantic Depth:** DINOv2 was chosen for its robust pre-training on massive datasets via self-supervised learning, enabling the extraction of rich semantic features and fine-grained visual details.
- **Task Suitability:** This architecture is considered ideal for lip-reading tasks, where distinguishing subtle visual cues (visemes) is critical for performance.
- **Efficiency:** The `vit_small_patch14` variant was utilized to maintain an optimal balance between high-performance feature extraction and manageable VRAM consumption within a 15 GB limit.

3.2. Methodology and Training Pipeline

A **Two-Phase Joint Training** strategy was implemented to overcome the performance plateau observed with strictly frozen features.

3.2.1. Phase 1: Temporal Head Initialization

- The DINOv2 backbone was kept **frozen** to preserve pre-trained spatial knowledge.
- Training was restricted to the **Temporal Head**, a Transformer-based sequence processor, using pre-extracted .pt features.
- This phase established a rapid temporal baseline without significant computational overhead.
- Phase 1 trained for 39 epochs.

3.2.2. Phase 2: Joint Fine-Tuning

- The final **4 blocks** of the DINOv2 backbone were unfrozen (p2_unfreeze_n: 4) and trained jointly with the Temporal Head.
- Videos were processed directly from .npz files, allowing the backbone to adapt its spatial perception specifically to Turkish phonemes and dynamic lip movements.
- Phase 2 trained for 8 epochs.

3.3. Fine-Tuning and Optimization Strategy

A specialized protocol was followed to address the performance bottleneck identified in initial experiments.

- **Block-Specific Unfreezing:** To prevent the degradation of robust DINOv2 features, only the final layers of the Vision Transformer (ViT) were set to a trainable state.
- **Differential Learning Rates:** Architectural stability was maintained by assigning a very low learning rate (**5e-7**) to the backbone and a relatively higher rate (**1e-5**) to the Temporal Head.
- **Joint Optimization:** Simultaneous optimization of the spatial extractor and temporal Transformer allowed the model to refine its focus on the lip region in coordination with sequence modeling.

3.4. Architectural Configuration

The final model integrates a high-performance visual backbone with a sequence-modeling head:

- **Backbone:** vit_small_patch14_dinov2 with an embedding dimension of 384
- **Temporal Head:** A Transformer architecture with a model dimension of 384 and 4 attention heads
- **Depth:** Consists of 4 Transformer layers with a feedforward network dimension of 1024
- **Dropout:** A 0.3 dropout rate was implemented within the Transformer layers to ensure robust feature learning

3.5. Hyperparameters and Implementation Details

- **Batch Configuration:** Training utilized a batch size of 64
- **Optimization:** The AdamW optimizer was used with an initial learning rate of 3e-5 and weight decay of 1e-4
- **Scheduling:** A ReduceLROnPlateau scheduler was employed (factor: 0.5, patience: 5 epochs)
- **Duration:** Training was scheduled for 50 epochs with early stopping after 10 epochs of stagnation
- **Efficiency:** Frames were processed in extract_chunk blocks of 8 to optimize memory usage
- **Regularization:** A label smoothing factor of 0.1 was applied

3.6. Performance Evaluation

The transition to joint fine-tuning resulted in a measurable performance breakthrough:

- **Accuracy:** Validation accuracy increased from a plateau of ~24% to 40.51%
- **Loss:** Validation loss was reduced from 3.68 to 2.77, indicating significantly higher model confidence
- **Recall@5:** A score of 69.13% was achieved, confirming that the correct word is identified within the top five candidates in the vast majority of cases

Phase 1 Results:

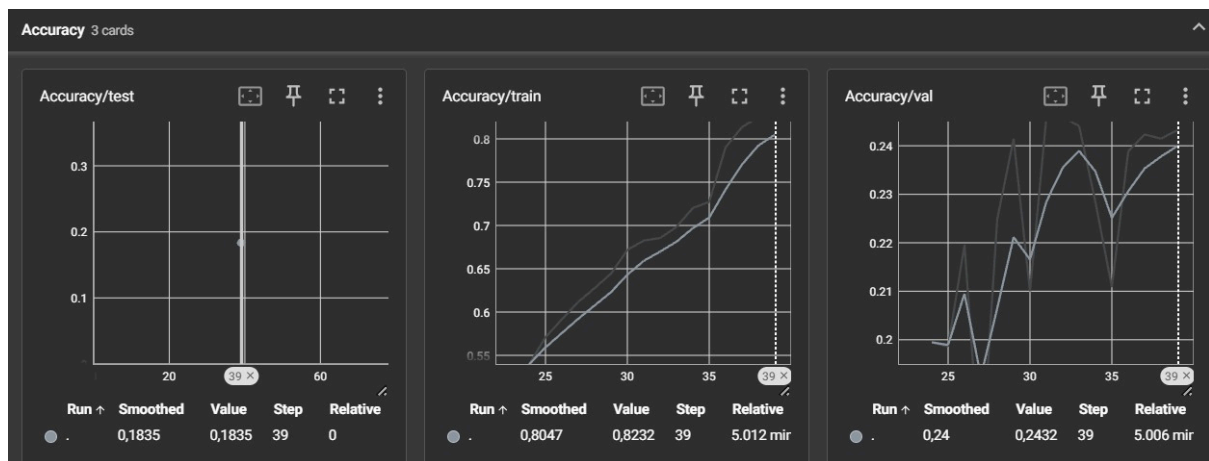


Figure 5: Phase 1 Accuracy Results

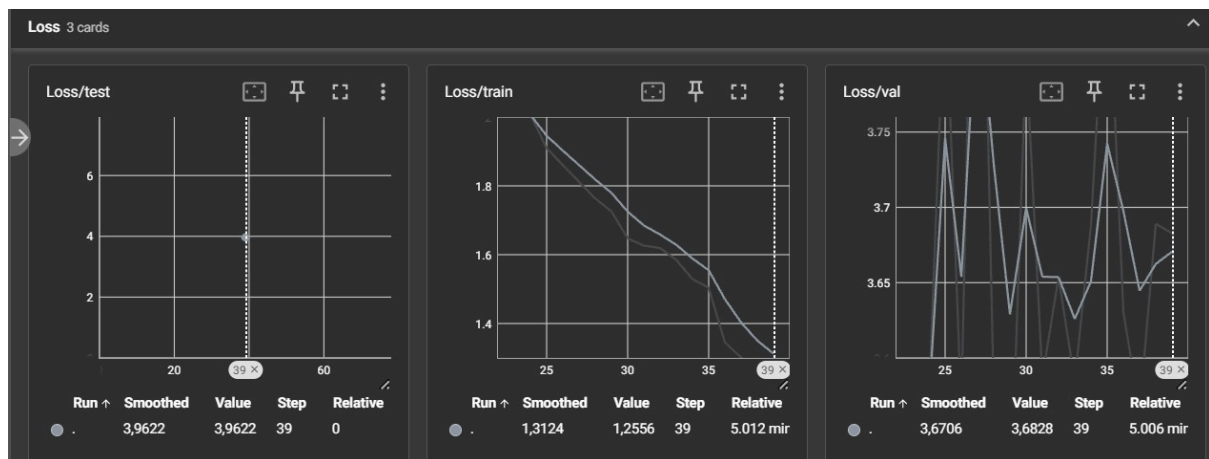


Figure 6: Phase 1 Loss Results

Phase 2 Results:

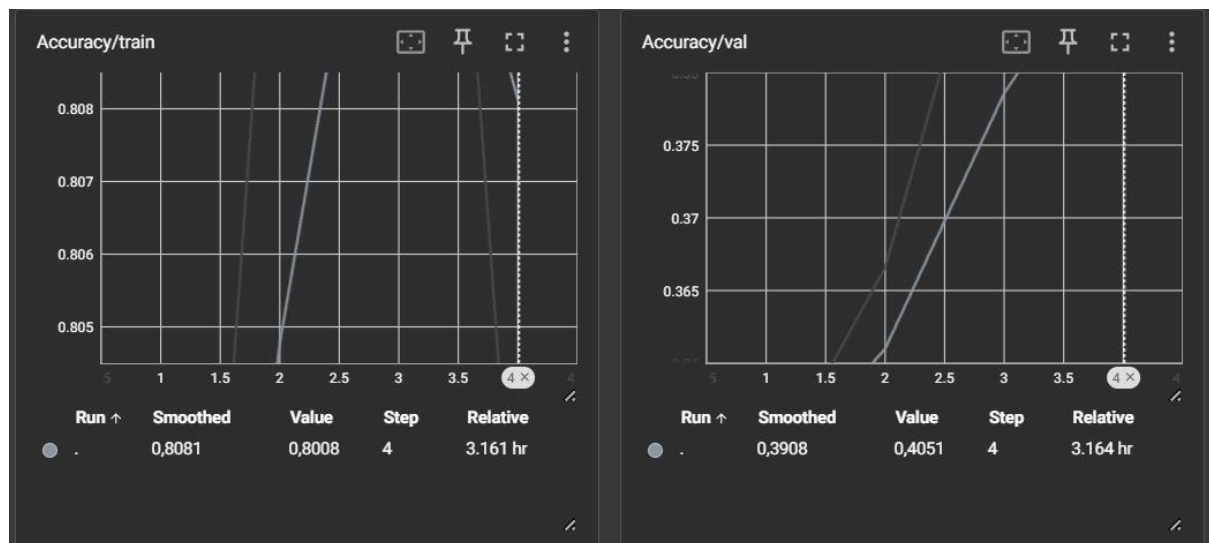


Figure 7: Last 4 Epochs of Phase 2 Accuracy Results

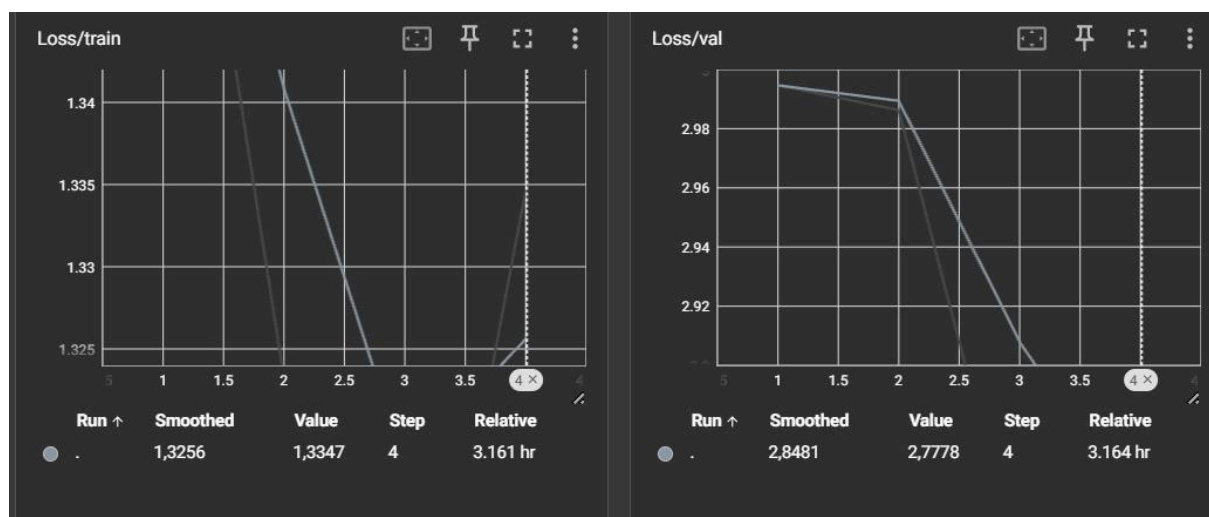


Figure 8: The Last 4 Epochs of Phase 2 Loss Results

Backbone: last 4/12 blocks unfrozen.

Phase 2 starting from scratch.

[P2 train] E001 | loss=1.3702 | acc=0.7906

[val] E001 | loss=2.9947 | acc=0.3517 | f1=0.3248 | wer=0.6483

✓ Best saved (E1 | loss=2.9947 | acc=0.3517)

[P2 train] E002 | loss=1.3232 | acc=0.8133

[val] E002 | loss=2.9863 | acc=0.3666 | f1=0.3370 | wer=0.6334

✓ Best saved (E2 | loss=2.9863 | acc=0.3666)

[P2 train] E003 | loss=1.2958 | acc=0.8235

[val] E003 | loss=2.8295 | acc=0.3955 | f1=0.3660 | wer=0.6045

✓ Best saved (E3 | loss=2.8295 | acc=0.3955)

[P2 train] E004 | loss=1.3347 | acc=0.8008

[val] E004 | loss=2.7778 | acc=0.4051 | f1=0.3732 | wer=0.5949

✓ Best saved (E4 | loss=2.7778 | acc=0.4051)

Figure 9: The Last 4 Epochs of Phase 2

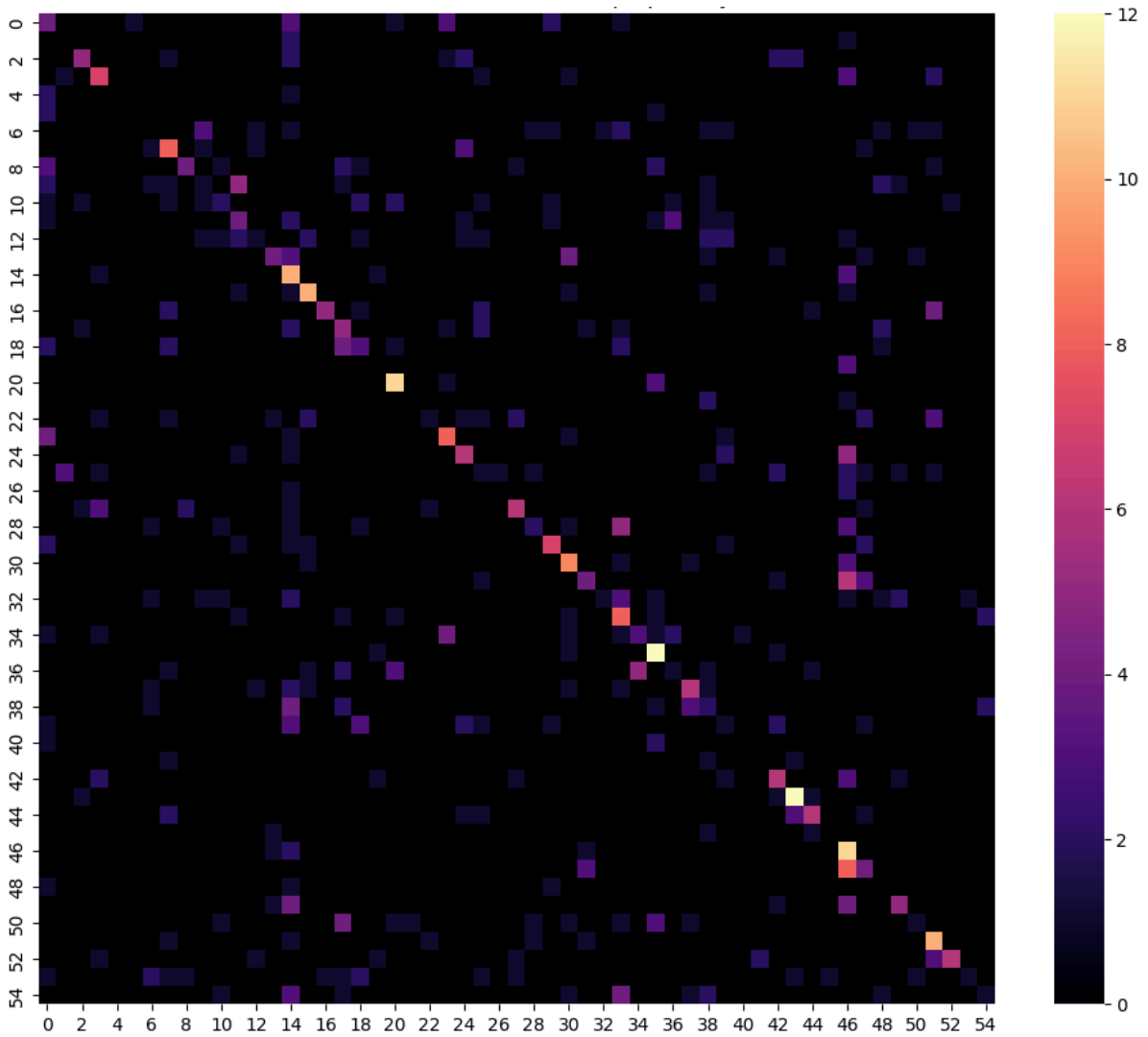


Figure 10: Confusion Matrix

4. Hybrid ConvNeXt + Transformer Model

4.1. Model Architecture

A Hybrid Spatio-Temporal Model was implemented for this system. The architecture integrates a ConvNeXt-Tiny spatial backbone with a 4-layer Transformer Encoder to process the sequential nature of lip movements.

4.2. Model Selection Rationale

- **ConvNeXt-Tiny (Spatial Encoder):** This modern convolutional architecture was selected to extract high-dimensional (768-dimensional) spatial features from each

frame. It offers the efficiency of CNNs with the advanced performance of Vision Transformers.

- **Transformer (Temporal Encoder):** Since lip reading relies on the chronological order of frames, a Transformer was implemented to capture long-range temporal dependencies. The Self-Attention mechanism allows the model to focus on critical frame transitions that define specific Turkish words.

4.3. Processing Pipeline

The system follows a streamlined pipeline, utilizing a custom-built and pre-processed dataset:

- **Data Loading:** Each video has been previously converted into a sequence of 96×96 resolution mouth-centered (ROI) images.
- **Sequence Standardization:** Sequences are standardized to a fixed length of 75 frames using padding or truncation.
- **Spatial Embedding:** Each 96×96 frame is processed by ConvNeXt-Tiny, converting visual data into a 768-dimensional vector.
- **Temporal Processing:** Positional encodings are added to these vectors, which are then processed by the Transformer layers to model the speech dynamics over time.
- **Classification:** Global average pooling is applied over the temporal dimension, followed by a final dense layer predicting one of the 55 Turkish word classes.

4.4. Training Stage 1: Frozen Backbone Initialization

During the initial phase of training, a "Frozen Backbone" strategy was implemented to establish a stable learning foundation.

- **Methodology:** For the first 5 epochs, the ConvNeXt-Tiny spatial backbone was frozen (`requires_grad = False`). Only the Transformer layers and the final classification head were trained.
- **Purpose:** This protected the sophisticated pre-trained weights of the ConvNeXt model from being corrupted by initially high error gradients.
- **Effect:** The model successfully focused on learning the temporal "rhythm" and sequence patterns of lip movements, creating a solid baseline before high-precision spatial tuning.

4.5. Training Stage 2: Strategic Refinement (Full Fine-Tuning)

Once stabilization was achieved, the training flow was manually intervened to initiate a deep **Refinement** phase.

- **Methodology (Unfreezing):** All model parameters were set to trainable, allowing the ConvNeXt backbone to shift from general object recognition to a specialized understanding of Turkish lip morphology (visemes).
- **Advanced Custom Augmentation:**
 - **Random Rotation (5°):** Simulates natural head tilts and perspective shifts.

- **Color Jitter (Brightness/Contrast/Saturation):** Simulates real-world environmental variability in lighting conditions.
- **Gaussian Blur (p=0.3):** Accounts for hardware and movement-related imperfections (camera focus loss, motion blur).
- **No Horizontal Flipping:** A critical design choice — in Turkish lip reading, the direction and asymmetrical nature of certain phonetic movements are vital. Flipping would create visually impossible visemes.

4.6. Evaluation Metrics

The evaluation process utilized a comprehensive set of metrics:

- **Precision & Recall:** Measures the model's exactness and completeness, helping identify if the model is biased toward certain common words.
- **F1-Score:** Selected as the primary balancing metric, especially important for the 55-class classification.
- **Top-K Accuracy (Top-1, 2, 3, 5):** Provides a better understanding of the model's performance given the inherent ambiguity of lip reading.
- **Validation & Test Accuracy:** Monitored to track generalization and ensure the refined model was not overfitting.

4.7. Experimental Results and Analysis

4.7.1. Global Performance Metrics: The 57.15% Macro F1-score confirms that the refinement process successfully taught the model to distinguish between classes with varying sample frequencies, ensuring high discriminative power across the entire dataset.

Metric	Score
Top-1 Accuracy	55.66% (0.56)
Macro Average F1-Score	57.15% (0.57)
Weighted Average F1-Score	56.00%

4.7.2. Validation Accuracy

The validation accuracy graph illustrates a significant performance surge during the strategic refinement phase, reaching a new record of **61.21%**.

```
📁 Epoch 8 kaydedildi: /content/checkpoints/refined_process/refined_epoch_8_acc_61.21.pt  
★ YENİ REKOR! %61.21 ile en iyi model güncellendi.
```

Figure 11: [Validation Accuracy Results]

4.7.3 Top-K Accuracy and User Experience

In visual speech recognition, Top-K metrics are often more representative of system utility than Top-1 accuracy alone:

- **Top-2 Accuracy:** 70.77%
- **Top-3 Accuracy:** 78.36%
- **Top-5 Accuracy:** 84.74%

The 84.74% Top-5 accuracy indicates that for almost 85% of the test samples, the correct word was included in the model's top 5 suggestions, making the system highly viable as an assistive technology.

```
🔍 Top-K analizi yapılıyor (K: [1, 2, 3, 5])...  
  
📊 --- TOP-K ACCURACY SONUÇLARI (Test Seti) ---  
🎯 Top-1 Accuracy: %55.66  
🎯 Top-2 Accuracy: %70.77  
🎯 Top-3 Accuracy: %78.36  
🎯 Top-5 Accuracy: %84.74
```

Figure 12:[Top-K accuracy results for K=[1, 2, 3, 5], showing that the correct gesture is within the model's top 5 predictions 84.74% of the time.]

4.7.4 Class-Specific Analysis and Success Cases

The system performs exceptionally well on words with clear morphological structures:

Top Performers:

- **"İstiyorum"**: Perfect Precision of 1.00 and F1-score of 0.83
- **"Bekliyorum"**: Very high Precision of 0.95 and F1-score of 0.86
- **"Açmak"**: Precision of 0.94 and F1-score of 0.75
- **"Yardımcı"**: Precision of 0.94 and F1-score of 0.81

High-Consistency Classes: Words like "Buzdolabı" (F1: 0.78), "Doktor" (F1: 0.82), and "Şimdi" (F1: 0.78) maintain high scores across diverse conversational topics.

--- REFINED MODEL CLASSIFICATION REPORT ---				
	precision	recall	f1-score	support
acıktım	0.61	0.48	0.54	23
açmak	0.94	0.62	0.75	24
anlıyorum	0.52	0.62	0.57	24
araba	0.20	0.38	0.26	24
arkadaş	0.67	0.67	0.67	24
aşağı	0.62	0.54	0.58	24
bebek	0.17	0.21	0.19	24
bekliyorum	0.95	0.79	0.86	24
ben	0.32	0.50	0.39	24
bir	0.36	0.42	0.38	24
biraz	0.86	0.25	0.39	24
bitti	0.19	0.21	0.20	24
biz	0.30	0.38	0.33	24
buzdolabı	0.82	0.75	0.78	24
doktor	0.80	0.83	0.82	24
domates	0.63	0.71	0.67	24
elbise	0.62	0.54	0.58	24
fakat	0.71	0.42	0.53	24
fenner	0.45	0.42	0.43	24
fotoğraf	0.78	0.75	0.77	24
geldim	0.37	0.54	0.44	24
güzel	0.68	0.54	0.60	24
hemen	0.58	0.58	0.58	24
herhangi	0.39	0.58	0.47	24
istiyorum	1.00	0.71	0.83	24
kapı	0.86	0.25	0.39	24
lütfe	0.59	0.79	0.68	24
merhaba	0.27	0.67	0.38	24
mumya	0.50	0.79	0.61	24
nasılsın	0.71	0.50	0.59	24
oyuncak	0.89	0.67	0.76	24

Figure 13: [Detailed classification report showing precision, recall, and F1-score for each word]

pamuk	0.39	0.75	0.51	24
papatya	0.80	0.50	0.62	24
postane	0.92	0.50	0.65	22
sandalye	0.79	0.46	0.58	24
selam	0.73	0.46	0.56	24
sen	0.60	0.25	0.35	24
sonra	0.52	0.54	0.53	24
susadım	0.81	0.54	0.65	24
sınıf	1.00	0.46	0.63	24
şey	0.73	0.67	0.70	24
şimdi	0.94	0.67	0.78	24
tamam	0.41	0.79	0.54	24
telefon	0.57	0.50	0.53	24
televizyon	0.74	0.58	0.65	24
eşekkürler	0.81	0.54	0.65	24
umut	0.81	0.71	0.76	24
vapur	0.69	0.75	0.72	24
vazgeçtim	0.84	0.67	0.74	24
vazo	0.65	0.54	0.59	24
ve	0.20	0.58	0.30	24
yapmak	0.62	0.33	0.43	24
yardımcı	0.94	0.71	0.81	24
yemek	0.45	0.38	0.41	24
yukarı	0.88	0.62	0.73	24
accuracy			0.56	1317
macro avg	0.64	0.56	0.57	1317
weighted avg	0.64	0.56	0.57	1317

Final Macro F1-Score: %57.15

Figure 14: [Detailed classification report showing precision, recall, and F1-score for each word achieving an overall macro F1-score of 57.15%.]

4.7.5 Qualitative Case Study: Top-5 Prediction Analysis

By examining the Top-5 prediction lists for words like Postane, Aşağı, İstiyorum, Bebek, and Domates:

- **High-Certainty Success (İstiyorum & Bebek):** "İstiyorum" is identified with 80.90% confidence, and "Bebek" with 59.54%. The significant probability gap between the first and second candidates proves that the model has learned highly distinct spatio-temporal "signatures" for these words.
- **Sequential Reliability (Postane & Aşağı):** The target word is successfully ranked within the Top-5. Even when the visual cues are subtle (such as the soft vowel transitions in "Aşağı"), the Transformer Encoder effectively utilizes sequence timing to keep the correct word as a high-probability candidate.
- **The Viseme Conflict (Domates vs. Fener):** The word "Domates" was predicted as "Fener" with 62.55% confidence. However, "Domates" remains in the Top-5 list. This confirms the model correctly identifies the viseme group but requires the Top-5 safety net due to near-identical visual phonetics.

🔍 5 adet video için Top-5 aday listesi:

🔍 Gerçek Kelime: **POSTANE**			
Sıra	Tahmin Edilen		Güven Oranı (Confidence)
1	postane	★	%69.28
2	domates		%6.42
3	şey		%3.94
4	bitti		%1.57
5	şimdi		%1.09

🔍 Gerçek Kelime: **AŞAĞI**			
Sıra	Tahmin Edilen		Güven Oranı (Confidence)
1	acıktım		%61.12
2	bir		%6.91
3	selam		%4.66
4	ben		%1.97
5	herhangi		%1.81

🔍 Gerçek Kelime: **İSTİYORUM**			
Sıra	Tahmin Edilen		Güven Oranı (Confidence)
1	istiyorum	★	%80.90
2	bekliyorum		%2.58
3	selam		%1.40
4	teşekkürler		%0.50
5	tamam		%0.50

Figure 15: [Top-5 Prediction For 5 words]

🔍 Gerçek Kelime: **BEBEK**			
Sıra	Tahmin Edilen	Güven Oranı (Confidence)	
1	bebek	★ %59.54	
2	papatya	%5.31	
3	bitti	%3.99	
4	tamam	%3.34	
5	domates	%1.76	
=====			
🔍 Gerçek Kelime: **DOMATES**			
Sıra	Tahmin Edilen	Güven Oranı (Confidence)	
1	fener	%62.55	
2	herhangi	%7.79	
3	arkadaş	%3.93	
4	fakat	%3.26	
5	araba	%1.14	
=====			

Figure 16: [Top-5 Prediction For 5 words]

4.7.6 Inference Analysis on Random Samples

To evaluate real-world performance, an inference test was conducted on randomly selected video samples from the test set. In this specific trial, the system successfully identified **5 out of 10 samples correctly**, achieving a **50.00% Top-1 accuracy** for this randomized batch.

🔍 10 adet rastgele test videosu işleniyor...		
GERÇEK KELİME	MODEL TAHMİNİ	DURUM
fakat	ve	❌ HATALI
oyuncak	fotoğraf	❌ HATALI
yukarı	şey	❌ HATALI
buzdolabı	buzdolabı	✅ DOĞRU
biz	biz	✅ DOĞRU
fakat	fakat	✅ DOĞRU
fotoğraf	fotoğraf	✅ DOĞRU
nasılsın	bitti	❌ HATALI
biz	ben	❌ HATALI
sandalye	sandalye	✅ DOĞRU

Figure 17: [Examples of model predictions on random test videos, highlighting successful classifications and common misidentifications]

4.7.7 Viseme Analysis and Error Patterns (Confusion Matrix)

The errors made by the model are not random but are deeply linked to the concept of visemes — different phonemes that look identical when spoken.



Figure 18: [Confusion Matrix]

- **Confusion Matrix Insights:** Clusters of confusion are visible between phonetically different but visually similar groups (e.g., the B-P-M or D-N-T groups).

4.7.8 Conclusion of the Refinement Impact

The manual transition from the Frozen stabilization phase to the Strategic Refinement phase transformed the system into a specialized Turkish lip-reading expert. The 57.15% Macro F1 and 84.74% Top-5 Accuracy prove that the Hybrid ConvNeXt-Transformer model is a robust architecture capable of overcoming the significant challenges of visual-only speech recognition in the Turkish language.

5. ViViT Model

5.1. Model Selection

In this study, a hybrid Video Vision Transformer (ViViT) architecture was adopted for the lip-reading task. The model utilizes a `vit_small_patch16_224` backbone as the spatial

feature extractor, coupled with a custom TransformerEncoder designed to capture temporal dynamics.

The selection of a ViT-based model was motivated by its superior capability to capture global contextual information compared to standard 2D CNNs. Lip-reading is inherently a sequential task where the temporal relationship between phonemes determines the meaning. This hybrid structure allows the model to effectively map spatio-temporal features to word-level classifications by leveraging pre-trained spatial representations and learning dependencies across frames via the transformer encoder.

5.2. Processing Pipeline

The data pipeline was engineered as an end-to-end framework to minimize latency and ensure consistency:

- **Preprocessing:** All input videos were standardized to a fixed frame rate of 25 FPS. Frames were converted to grayscale (in_chans=1) and resized to 224 × 224 pixels to align with the ViT backbone requirements.
- **Sequential Alignment:** Sequences were fixed to a length of 50 frames. Videos shorter than this duration were padded with zeros, while longer sequences were centered and cropped.
- **Feature Extraction:** Each frame is processed through the ViT backbone, generating a latent feature vector representation.
- **Temporal Encoding:** These vectors are passed through a three-layer TransformerEncoder (nhead=6, d_model=384) to model the temporal progression of lip movements.
- **Classification:** The output of the temporal encoder is globally averaged and passed to a linear classification head for final label prediction.

5.3. Training Strategy: Augmentation and Fine-Tuning

To enhance model robustness and prevent overfitting, the following strategies were implemented:

- **Data Augmentation:** Techniques including random brightness adjustments and slight affine rotations were applied during training via the LipReadingDataset class. These simulate real-world variability in lighting and head positioning.
- **Fine-Tuning:** The ViT backbone, originally pre-trained on ImageNet, was fine-tuned for the lip-reading domain. By utilizing pre-trained weights, the model leverages established spatial patterns while adapting to the fine-grained, dynamic features necessary for identifying specific speech movements. This transfer learning approach significantly accelerated convergence compared to training from scratch.

5.4. Evaluation Framework

The performance of the model was evaluated using Precision, Recall, F1-Score, and overall Accuracy.

- **Precision and Recall** were prioritized to diagnose specific class failures, indicating whether the model was over-predicting specific labels or missing instances entirely.
- **F1-Score** was utilized as the primary harmonic metric to ensure a balance between Precision and Recall.
- **Confusion Matrix Analysis:** The confusion matrix was visualized to identify systematic errors and determine which phonetic classes were most frequently misclassified.

```
>> Epoch 5 | Train Loss: 0.6311, Acc: 80.50% | Val Loss: 2.0936, Acc: 45.43%
Yeni en iyi model kaydedildi!
Epoch 6/20 [Train]: 100%|██████████| 1191/1191 [04:22<00:00, 4.55it/s, loss=0.0929, lr=0.0]
Epoch 6/20 [Val]: 100%|██████████| 144/144 [00:18<00:00, 7.96it/s]
```

Figure 19: [Screenshot of the last epoch]

```
🚀 Test aşaması başladı...
*** Test: 100%|██████████| 165/165 [18:41<00:00, 6.79s/it]

--- CLASSIFICATION REPORT ---
      precision    recall  f1-score   support

     0         0.48      0.42      0.44         24
     1         0.28      0.38      0.32         24
     2         0.50      0.33      0.40         24
     3         0.38      0.38      0.38         24
     4         0.55      0.26      0.35         23
     5         0.58      0.46      0.51         24
     6         0.23      0.25      0.24         24
     7         0.48      0.62      0.55         24
     8         0.62      0.33      0.43         24
     9         0.21      0.54      0.31         24
    10         0.31      0.21      0.25         24
    11         0.09      0.12      0.10         24
    12         0.16      0.21      0.18         24
    13         0.33      0.46      0.39         24
    14         1.00      0.50      0.67         24
    15         0.75      0.38      0.50         24
    16         0.19      0.38      0.25         24
    17         0.69      0.46      0.55         24
    18         0.57      0.17      0.26         24
    19         0.57      0.17      0.26         24
    20         0.83      0.42      0.56         24
    21         0.54      0.58      0.56         24
    22         0.22      0.08      0.12         24
    23         0.29      0.17      0.21         24
    24         0.15      0.79      0.25         24
    25         0.15      0.21      0.17         24
    26         0.60      0.38      0.46         24
    27         0.53      0.42      0.47         24
    28         0.81      0.54      0.65         24
    29         0.37      0.29      0.33         24
    30         1.00      0.17      0.29         24
```

Figure 20: [Classification report]

30	1.00	0.17	0.29	24
31	0.52	0.46	0.49	24
32	0.43	0.25	0.32	24
33	0.57	0.50	0.53	24
34	0.13	0.25	0.17	24
35	0.45	0.38	0.41	24
36	0.67	0.25	0.36	24
37	0.62	0.21	0.31	24
38	0.34	0.50	0.41	24
39	0.12	0.33	0.17	24
40	0.33	0.29	0.31	24
41	0.27	0.38	0.32	24
42	0.47	0.38	0.42	24
43	0.90	0.38	0.53	24
44	0.44	0.46	0.45	24
45	0.50	0.54	0.52	24
46	0.44	0.96	0.61	24
47	0.50	0.62	0.56	24
48	1.00	0.33	0.50	24
49	0.38	0.33	0.36	24
50	0.80	0.67	0.73	24
51	0.57	0.17	0.26	24
52	0.61	0.46	0.52	24
53	0.22	0.08	0.12	24
54	0.58	0.29	0.39	24
accuracy			0.37	1319
macro avg	0.48	0.37	0.38	1319
weighted avg	0.48	0.37	0.38	1319

Figure 21: [Classification Report]

Conclusion

This study presented a comprehensive evaluation of five deep learning architectures for Turkish word-level lip reading on a 55-class visual speech dataset. The experimental results demonstrate that hybrid architectures combining convolutional feature extractors with Transformer-based temporal encoders achieve the strongest performance. In particular, the 3D CNN + Conformer model attained the highest validation accuracy of 80.67% and an F1-score of 80.81%, establishing it as the most effective architecture for this task.

The ResNet18 + 3D CNN + Transformer model achieved a SOTA validation accuracy of 49.70%, demonstrating competitive performance despite the increased complexity introduced by the dual-stream design. The DINOv2 (ViT-Small) model, leveraging large-scale self-supervised pre-training, reached 40.51% validation accuracy after a two-phase joint fine-tuning strategy, highlighting the potential of foundation models for lip

reading tasks. The Hybrid ConvNeXt + Transformer model achieved a 61.21% validation accuracy and an 84.74% Top-5 accuracy, proving the value of modern convolutional backbones combined with Transformer temporal modeling. Finally, the ViViT model attained a 45.43% validation accuracy, demonstrating the applicability of video-oriented Vision Transformers for this problem.

Across all models, a recurring challenge is the visual ambiguity of homophenes — phonemes that are visually identical despite being acoustically distinct. Future work should investigate multi-modal fusion approaches incorporating audio signals, larger and more diverse Turkish lip reading datasets, and the integration of language model decoders to resolve phonetic ambiguity at the word and sentence level. The findings of this study provide a strong baseline for Turkish visual speech recognition research and lay the groundwork for real-world assistive communication systems.